The Blackwell–Dubins Theorem on the Merging of Opinions: Discrete and Continuous Time

Analytical Exposition

Contents

1 Introduction **2**

2 Setup and Notation **3**
- 2.1 The canonical sequence space . . . 3
- 2.2 Absolute continuity . . . 3
- 2.3 Total variation distance . . . 3
- 2.4 Conditional distributions and the merging question . . . 4

3 The Likelihood-Ratio Martingale **4**
- 3.1 The Radon–Nikodym process . . . 4
- 3.2 The Bayes formula for conditional posteriors . . . 5

4 The Blackwell–Dubins Theorem: Discrete Time **5**
- 4.1 Statement . . . 5
- 4.2 Proof . . . 5
- 4.3 Tightness and the role of mutual absolute continuity . . . 6

5 Extension to Continuous Time on $[0, +\infty)$ **6**
- 5.1 Filtered probability spaces and local martingales . . . 6
- 5.2 The likelihood-ratio process via Girsanov's theorem . . . 7
- 5.3 Global vs. local absolute continuity on $[0, +\infty)$ . . . 7
- 5.4 The Blackwell–Dubins theorem in continuous time . . . 8
- 5.5 The role of the energy condition on path space . . . 8

6 Kakutani's Theorem and the Absolute-Continuity Condition **9**
- 6.1 Infinite product measures . . . 9
- 6.2 Implication for merging . . . 10

7 Applications **10**
7.1 Bayesian learning 10
7.2 Rational expectations and the common prior assumption 10
7.3 Signal extraction and diffusion models 10
7.4 Ergodic Markov chains 11

8 Related Results and Extensions **11**
8.1 Lévy's zero-one law and Doob's consistency theorem 11
8.2 Diaconis–Freedman and robustness of Bayesian consistency 12
8.3 Rate of merging 12
8.4 Merging without absolute continuity 12
8.5 Summary of the main logical dependencies 12

1 Introduction

A foundational question in Bayesian epistemology and statistical decision theory is: if two agents hold different prior beliefs about a stochastic process but observe the same data indefinitely, will their probability assessments eventually converge? Blackwell and Dubins [1962] give a precise and general affirmative answer under the single hypothesis that the two priors are mutually absolutely continuous. Their result is now called the **merging-of-opinions theorem**, and it occupies a central place in Bayesian learning, rational expectations theory, and the ergodic theory of Markov chains.

The theorem's content can be stated informally as follows. Let P and Q be two probability measures on the space of infinite sequences (or infinite paths) of observations. If P is absolutely continuous with respect to Q, then—as the observed history grows without bound—the *conditional* distributions of the future path assigned by P and Q merge in total variation with probability one under both measures. No matter how different the two agents' initial beliefs were, they will, almost surely, make identical probability assessments about all future events once enough data have been observed.

Section 2 develops the measurable-space framework and recalls the total variation metric. Section 3 introduces the Radon–Nikodym likelihood-ratio process and its martingale properties. Section 4 states and proves the Blackwell–Dubins theorem in discrete time. Section 5 extends the analysis to continuous time on $[0, +\infty)$, where the Girsanov–Cameron–Martin theorem plays the role of the Radon–Nikodym derivative. Section 6 discusses Kakutani's dichotomy, which characterises exactly when mutual absolute continuity holds for product measures, and hence when merging is guaranteed. Section 7 discusses applications to Bayesian learning, rational expectations, and the ergodic theory of diffusions. Section 8 collects related results and extensions.

2 Setup and Notation

2.1 The canonical sequence space

Let $(S, \mathscr{S})$ be a measurable space (the *signal space*). Set $\Omega = S^{\mathbb{N}}$, the space of all sequences $\omega = (x_1, x_2, \ldots)$ with $x_n \in S$. Equip Ω with the product σ-algebra $\mathscr{F} = \mathscr{S}^{\otimes \mathbb{N}}$.

For each $n \geq 1$ define the finite-horizon σ-algebra

$$\mathscr{F}_n = \sigma(x_1, \ldots, x_n),$$

so that $\mathscr{F}_1 \subseteq \mathscr{F}_2 \subseteq \cdots \subseteq \mathscr{F}$. Let $\mathscr{F}_\infty = \sigma\big(\bigcup_{n \geq 1} \mathscr{F}_n\big)$. When S is a Polish space and $\mathscr{S}$ is the Borel σ-algebra, $\mathscr{F}_\infty = \mathscr{F}$.

Let P and Q denote two probability measures on $(\Omega, \mathscr{F})$. Write $P_n = P|_{\mathscr{F}_n}$ and $Q_n = Q|_{\mathscr{F}_n}$ for their restrictions to the history up to time n.

2.2 Absolute continuity

Definition 2.1 (Absolute continuity on a filtered space)**.** *P is **absolutely continuous** with respect to Q, written $P \ll Q$, if for every $A \in \mathscr{F}$, $Q(A) = 0$ implies $P(A) = 0$. They are **mutually absolutely continuous** (or* equivalent*), written $P \sim Q$, if both $P \ll Q$ and $Q \ll P$.*

*P is **locally absolutely continuous** with respect to Q if $P_n \ll Q_n$ for every $n \geq 1$; global absolute continuity $P \ll Q$ implies local absolute continuity but is strictly stronger.*

2.3 Total variation distance

For two probability measures μ and ν on a measurable space $(E, \mathscr{E})$, the **total variation distance** is

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{A \in \mathscr{E}} |\mu(A) - \nu(A)| = \tfrac{1}{2} \int_E \left| \frac{d\mu}{d\lambda} - \frac{d\nu}{d\lambda} \right| d\lambda,$$

where λ is any common dominating measure. Equivalently,

$$\|\mu - \nu\|_{\mathrm{TV}} = 1 - \int_E \min\left(\frac{d\mu}{d\lambda}, \frac{d\nu}{d\lambda}\right) d\lambda, \tag{1}$$

which takes values in $[0, 1]$. When $\mu \ll \nu$ with $f = d\mu/d\nu$,

$$\|\mu - \nu\|_{\mathrm{TV}} = \mathbb{E}_\nu[(f-1)^+] = \mathbb{E}_\nu[(1-f)^+] = 1 - \mathbb{E}_\nu[\min(f, 1)]. \tag{2}$$

2.4 Conditional distributions and the merging question

The Blackwell–Dubins theorem concerns the conditional distributions of the *future* given the *past*. For each n and each ω, let $P(\cdot \mid \mathscr{F}_n)(\omega)$ denote the conditional probability measure on $(\Omega, \mathscr{F})$ induced by P given the event $\{x_1 = \omega_1, \ldots, x_n = \omega_n\}$. Define analogously $Q(\cdot \mid \mathscr{F}_n)(\omega)$.

Definition 2.2 (Merging in total variation)**.** *P and Q* ***merge in total variation*** *if*

$$\|P(\cdot \mid \mathscr{F}_n) - Q(\cdot \mid \mathscr{F}_n)\|_{\mathrm{TV}} \to 0 \quad \textit{almost surely.}$$

The qualifier "almost surely" is with respect to the measure under which the convergence is asserted, typically Q (and hence also P when $P \ll Q$).

3 The Likelihood-Ratio Martingale

3.1 The Radon–Nikodym process

Since $P \ll Q$ implies $P_n \ll Q_n$ for every n, the **Radon–Nikodym derivative** (likelihood ratio) at stage n is

$$Z_n = \frac{dP_n}{dQ_n}, \qquad Z_n \geq 0 \;\; Q\text{-a.s.}, \qquad \mathbb{E}_Q[Z_n] = 1.$$

The global condition $P \ll Q$ guarantees that $Z = dP/dQ$ exists on $(\Omega, \mathscr{F})$, and

$$Z_n = \mathbb{E}[\, Z \mid \mathscr{F}_n \,] \quad Q\text{-a.s.} \tag{3}$$

That is, $\{Z_n, \mathscr{F}_n\}_{n \geq 1}$ is a **non-negative, uniformly integrable Q-martingale**.

Lemma 3.1 (Martingale convergence)**.** *The likelihood ratio process $\{Z_n\}$ satisfies:*

(i) $Z_n \to Z_\infty$ *Q-a.s. as $n \to \infty$.*

(ii) $Z_\infty = \mathbb{E}[\, Z \mid \mathscr{F}_\infty \,]$ *Q-a.s.*

(iii) $Z_n \to Z_\infty$ *in $L^1(Q)$, i.e., $\mathbb{E}_Q[|Z_n - Z_\infty|] \to 0$.*

(iv) When $\mathscr{F}_\infty = \mathscr{F}$, $Z_\infty = Z$ Q-a.s., and the L^1 convergence is a consequence of uniform integrability.

Proof. Parts (i) and (ii) follow from the almost-sure martingale convergence theorem [Doob, 1953], since $\{Z_n\}$ is a non-negative martingale and hence bounded in $L^1(Q)$. Uniform integrability of $\{Z_n\}$ follows from $Z \in L^1(Q)$ (via (3) and Jensen's inequality), which gives (iii). Part (iv) is the identification of the a.s. limit with the terminal Radon–Nikodym derivative. □

3.2 The Bayes formula for conditional posteriors

The following identity relates the conditional measures to Z_n:

Lemma 3.2 (Conditional Bayes formula)**.** *On the set $\{Z_n > 0\}$, for any $A \in \mathscr{F}$,*

$$P(A \mid \mathscr{F}_n) = \frac{\mathbb{E}[\, Z_\infty \cdot \mathbf{1}_A \mid \mathscr{F}_n \,]}{Z_n} \quad Q\text{-a.s.} \tag{4}$$

Equivalently, the Radon–Nikodym derivative of $P(\cdot \mid \mathscr{F}_n)$ with respect to $Q(\cdot \mid \mathscr{F}_n)$ on $\mathscr{F}$ is

$$\frac{dP(\cdot \mid \mathscr{F}_n)}{dQ(\cdot \mid \mathscr{F}_n)} = \frac{Z_\infty}{Z_n} \quad Q\text{-a.s. on } \{Z_n > 0\}. \tag{5}$$

Proof. Equation (4) is an application of the abstract Bayes theorem: for $B \in \mathscr{F}_n$ and $A \in \mathscr{F}$,

$$P(A \cap B) = \int_B \mathbb{E}[\, Z_\infty \mathbf{1}_A \mid \mathscr{F}_n \,]\, dQ = \int_B \frac{\mathbb{E}[\, Z_\infty \mathbf{1}_A \mid \mathscr{F}_n \,]}{Z_n} Z_n \, dQ = \int_B \frac{\mathbb{E}[\, Z_\infty \mathbf{1}_A \mid \mathscr{F}_n \,]}{Z_n} dP_n,$$

which identifies (4). Equation (5) follows by applying (2) to the ratio Z_∞ / Z_n. □

4 The Blackwell–Dubins Theorem: Discrete Time

4.1 Statement

Theorem 4.1 (Blackwell–Dubins, 1962)**.** *Let P and Q be probability measures on $(\Omega, \mathscr{F})$ with $P \ll Q$. Let*

$$d_n = \| P(\cdot \mid \mathscr{F}_n) - Q(\cdot \mid \mathscr{F}_n) \|_{\mathrm{TV}} \,.$$

Then $d_n \to 0$ almost surely under Q (and hence also under P).

4.2 Proof

The proof combines three ingredients: (a) an expression for d_n in terms of the likelihood-ratio process; (b) the supermartingale property of $\{d_n\}$; and (c) L^1 convergence of the likelihood ratio.

Step 1: Expression for d_n. By Lemma 3.2, on $\{Z_n > 0\}$ the conditional measure $P(\cdot \mid \mathscr{F}_n)$ is absolutely continuous with respect to $Q(\cdot \mid \mathscr{F}_n)$ with Radon–Nikodym derivative Z_∞ / Z_n. Applying (2) gives

$$d_n = 1 - \mathbb{E}\left[\min\left(\frac{Z_\infty}{Z_n}, 1 \right) \cdot Z_n \,\middle|\, \mathscr{F}_n \right]_Q = \mathbb{E}\left[\left(1 - \frac{Z_\infty}{Z_n} \right)^+ \,\middle|\, \mathscr{F}_n \right]_{Q(\cdot \mid \mathscr{F}_n)}. \tag{6}$$

Equivalently, after multiplying through by Z_n and taking the Q-expectation:

$$2\, d_n\, Z_n = \mathbb{E}[\, |Z_\infty - Z_n| \mid \mathscr{F}_n \,]_Q \quad Q\text{-a.s.} \tag{7}$$

Step 2: $\{d_n\}$ is a Q-supermartingale. Conditioning on more information cannot increase the difficulty of distinguishing two hypotheses; it can only reduce it. Formally, for any coupling of $P(\cdot|\mathscr{F}_n)$ and $Q(\cdot|\mathscr{F}_n)$,

$$d_n = \|P(\cdot|\mathscr{F}_n) - Q(\cdot|\mathscr{F}_n)\|_{\mathrm{TV}} \geq \|P(\cdot|\mathscr{F}_{n+1}) - Q(\cdot|\mathscr{F}_{n+1})\|_{\mathrm{TV}} \quad Q\text{-a.s.}$$

More precisely, since $P(\cdot|\mathscr{F}_n) = \mathbb{E}[\, P(\cdot|\mathscr{F}_{n+1}) \,|\, \mathscr{F}_n \,]$ and the TV norm is convex,

$$\mathbb{E}[\, d_{n+1} \,|\, \mathscr{F}_n \,] \leq d_n \quad Q\text{-a.s.} \tag{8}$$

Thus $\{d_n, \mathscr{F}_n\}$ is a non-negative Q-supermartingale bounded in $[0,1]$. By the supermartingale convergence theorem [Doob, 1953], $d_n \to d_\infty$ Q-almost surely for some $[0,1]$-valued random variable d_∞.

Step 3: The limit is zero. From (7) and the tower property,

$$\mathbb{E}_Q[d_n] \leq \tfrac{1}{2}\mathbb{E}_Q[|Z_\infty - Z_n|] \to 0$$

by the L^1 convergence in Lemma 3.1(iii). Hence $d_n \to 0$ in $L^1(Q)$. Since $d_n \to d_\infty$ Q-a.s. and $d_n \to 0$ in probability under Q, the two limits must coincide: $d_\infty = 0$ Q-a.s.

Since $P \ll Q$, every Q-null set is also P-null, so $d_n \to 0$ P-a.s. as well. □

4.3 Tightness and the role of mutual absolute continuity

Remark 4.2 (One-sided vs. mutual absolute continuity)**.** *The theorem requires only $P \ll Q$; it does not require $Q \ll P$. Under the one-sided condition:*

- *Merging holds Q-a.s. (and hence P-a.s. since $P \ll Q$).*
- *If additionally $Q \ll P$ (i.e., $P \sim Q$), merging holds a.s. under both measures, and both agents learn to agree.*

Remark 4.3 (Tightness)**.** *The theorem is tight in the following sense. If $P \perp Q$ (mutual singularity), then there exists a set A with $P(A) = 1$ and $Q(A) = 0$. By Lévy's zero-one law, $Q(A|\mathscr{F}_n) \to 0$ and $P(A|\mathscr{F}_n) \to 1$ almost surely, so $d_n \to 1$ rather than zero. The dichotomy between $P \ll Q$ and $P \perp Q$ is thus qualitatively sharp.*

5 Extension to Continuous Time on $[0, +\infty)$

5.1 Filtered probability spaces and local martingales

Let $(\Omega, \mathscr{F}, \{\mathscr{F}_t\}_{t\geq 0})$ be a filtered measurable space satisfying the usual conditions (right-continuity of $\{\mathscr{F}_t\}$ and completeness of $\mathscr{F}_0$). Let P and Q be two probability measures on $(\Omega, \mathscr{F}_\infty)$ where $\mathscr{F}_\infty = \sigma(\bigcup_{t\geq 0} \mathscr{F}_t)$.

Denote by $P_t = P|_{\mathscr{F}_t}$ and $Q_t = Q|_{\mathscr{F}_t}$ the restrictions to the filtration at time t.

Definition 5.1 (Local absolute continuity)**.** *P is **locally absolutely continuous** with respect to Q if $P_t \ll Q_t$ for all $t < \infty$. When additionally $P \ll Q$ on $\mathscr{F}_\infty$, one says P is **globally absolutely continuous**.*

Note that local absolute continuity does *not* imply global absolute continuity. The gap between them is a key subtlety of the continuous-time theory.

5.2 The likelihood-ratio process via Girsanov's theorem

On the canonical Wiener space, $(\Omega, \mathscr{F}, Q)$ is the Wiener measure so that the coordinate process $W = \{W_t\}$ is a standard Brownian motion. Suppose that under P the process W has an additional drift: P-almost surely,

$$W_t = \widetilde{W}_t + \int_0^t \theta_s \, ds,$$

where $\widetilde{W}$ is a P-Brownian motion and $\{\theta_s\}$ is an $\{\mathscr{F}_s\}$-adapted process. The **Girsanov–Cameron–Martin theorem** [Girsanov, 1960] states that, if the stochastic exponential

$$Z_t = \mathcal{E}\left(\int_0^\cdot \theta_s \, dW_s\right)_t = \exp\left(\int_0^t \theta_s \, dW_s - \frac{1}{2}\int_0^t \theta_s^2 \, ds\right) \tag{9}$$

is a Q-martingale (not merely a local martingale), then $P_t \ll Q_t$ for all t with $dP_t/dQ_t = Z_t$.

The process Z_t is always a non-negative Q-local martingale; it is a true martingale if and only if $\mathbb{E}_Q[Z_t] = 1$ for all t. Sufficient conditions for this are:

- **Novikov's condition** [Novikov, 1972]: $\mathbb{E}_Q\left[\exp\left(\frac{1}{2}\int_0^T \theta_s^2 \, ds\right)\right] < \infty$ for all T.
- **Kazamaki's condition**: the process $\exp\left(\frac{1}{2}\int_0^\cdot \theta_s \, dW_s\right)$ is a uniformly integrable Q-submartingale.

5.3 Global vs. local absolute continuity on $[0, +\infty)$

A central issue on the infinite time horizon is that local absolute continuity (Z_t a true martingale for all t) does *not* ensure $P \ll Q$ on $\mathscr{F}_\infty$.

Proposition 5.2 (Dichotomy at infinity)**.** *Suppose Z_t defined by (9) is a true Q-martingale on $[0, T]$ for every finite T. Then:*

(i) $Z_t \to Z_\infty$ Q-a.s. as $t \to \infty$.

(ii) If $\{Z_t\}$ is uniformly integrable over $[0, \infty)$, then $P \ll Q$ on $\mathscr{F}_\infty$ with $dP/dQ = Z_\infty > 0$ P-a.s.

(iii) If $\{Z_t\}$ is not *uniformly integrable, then $Z_\infty = 0$ Q-a.s., and $P \perp Q$ on $\mathscr{F}_\infty$.*

In case (iii), P and Q become mutually singular at the terminal σ-algebra even though each finite-horizon restriction satisfies $P_t \ll Q_t$.

Proof. Since $Z_t \geq 0$ is a Q-martingale, $Z_t \to Z_\infty$ Q-a.s. by the almost-sure convergence theorem. In case (ii), uniform integrability gives $L^1(Q)$ convergence, hence $\mathbb{E}_Q[Z_\infty] = \lim_t \mathbb{E}_Q[Z_t] = 1$, which is the condition for $P \ll Q$. In case (iii), $\mathbb{E}_Q[Z_\infty] = 0$ forces $Z_\infty = 0$ Q-a.s.; one then constructs a set of full P-measure and zero Q-measure using the set $\{Z_\infty = 0\}$. □

Remark 5.3 (Sufficient condition via energy)**.** *A sufficient condition for uniform integrability of Z_t is*

$$\int_0^\infty \theta_s^2 \, ds < \infty \quad Q\text{-a.s.} \tag{10}$$

Intuitively, (10) *says the two measures differ only by a finite total "information" over the infinite horizon. When the drift θ is stationary and non-zero, condition* (10) *fails, P and Q are mutually singular at infinity, and there is no merging.*

5.4 The Blackwell–Dubins theorem in continuous time

Theorem 5.4 (Merging on $[0, +\infty)$)**.** *Let P and Q be probability measures on $(\Omega, \mathscr{F}_\infty)$ with $P \ll Q$. Let $Z_t = dP_t/dQ_t$ be the likelihood-ratio process. Define*

$$d_t = \|P(\cdot \mid \mathscr{F}_t) - Q(\cdot \mid \mathscr{F}_t)\|_{\mathrm{TV}} \, .$$

Then $d_t \to 0$ almost surely under Q (and hence under P) as $t \to +\infty$.

Proof sketch. The argument parallels the discrete-time proof. Under $P \ll Q$, the process $Z_t = \mathbb{E}[\, Z_\infty \mid \mathscr{F}_t \,]_Q$ is a uniformly integrable Q-martingale and $Z_t \to Z_\infty$ in $L^1(Q)$. The continuous-time analogue of (8) holds by the same conditional-convexity argument: $\{d_t, \mathscr{F}_t\}$ is a non-negative Q-supermartingale bounded in $[0, 1]$, hence $d_t \to d_\infty$ Q-a.s. The L^1 bound $\mathbb{E}_Q[d_t] \leq \frac{1}{2}\mathbb{E}_Q[|Z_t - Z_\infty|] \to 0$ forces $d_\infty = 0$ Q-a.s. □

5.5 The role of the energy condition on path space

On the Wiener space with drift θ, the condition $P \ll Q$ is equivalent to

$$\mathbb{E}_Q\left[\exp\left(\frac{1}{2}\int_0^\infty \theta_s^2 \, ds\right)\right] < \infty \tag{11}$$

(Novikov on the infinite horizon), or equivalently to the stochastic exponential Z_t being uniformly integrable.

When (11) holds:

- Both P and Q assign probability one to paths on which the empirical average $\frac{1}{t}\int_0^t \theta_s\, ds \to 0$ as $t \to \infty$.
- The observed increment $dW_t = dW_t^Q$ distinguishes the drift only locally, and the accumulated evidence is not sufficient to render the two measures singular.
- By Theorem 5.4, the conditional distributions of the future path merge.

When (11) fails (e.g., $\theta_s = \theta_0 > 0$ constant), the sample-path average $\frac{1}{t}\int_0^t dW_s$ satisfies a strong law under each measure that distinguishes them with probability one: P and Q are mutually singular, and the conditional distributions diverge rather than merge.

6 Kakutani's Theorem and the Absolute-Continuity Condition

A natural question is: when does the hypothesis of Theorem 4.1 hold? For product measures, the answer is given by a classical theorem of Kakutani [1948].

6.1 Infinite product measures

Let $P = \bigotimes_{n=1}^{\infty} P_n$ and $Q = \bigotimes_{n=1}^{\infty} Q_n$ be infinite product measures, where P_n and Q_n are probability measures on $(S, \mathscr{S})$. The **Hellinger affinity** between P_n and Q_n is

$$\rho_n = \int_S \sqrt{\frac{dP_n}{d\lambda} \cdot \frac{dQ_n}{d\lambda}}\, d\lambda,$$

where λ is any common dominating measure; $\rho_n \in (0, 1]$ with $\rho_n = 1$ iff $P_n = Q_n$.

Theorem 6.1 (Kakutani's Dichotomy, 1948)**.** *For infinite product measures:*

$$P \sim Q \quad \Longleftrightarrow \quad \prod_{n=1}^{\infty} \rho_n > 0 \quad \Longleftrightarrow \quad \sum_{n=1}^{\infty} (1 - \rho_n) < \infty.$$

If $\prod_{n=1}^{\infty} \rho_n = 0$, then $P \perp Q$. There is no intermediate case: either the measures are equivalent or mutually singular.

Proof sketch. The likelihood ratio $Z_n = \prod_{k=1}^{n} (dP_k/dQ_k)$ is a Q-martingale. The product $\mathbb{E}_Q[\sqrt{Z_n}] = \prod_{k=1}^{n} \rho_k$ is the Hellinger integral. If $\prod \rho_k > 0$, then Z_n is bounded in $L^{1/2}$, which implies uniform integrability in L^1 and hence $P \ll Q$. If $\prod \rho_k = 0$, then $Z_n \to 0$ in $L^{1/2}$, which forces $Z_\infty = 0$ Q-a.s., establishing $P \perp Q$. $\square$

6.2 Implication for merging

Kakutani's theorem identifies precisely the regime in which Blackwell–Dubins applies to i.i.d.-type sequences. If the one-period measures P_n and Q_n are "close" in the sense that $\sum_n (1 - \rho_n) < \infty$, then merging is guaranteed. If they differ by a fixed total variation amount at every period (e.g., P_n and Q_n are distinct fixed measures independent of n), the Hellinger product converges to zero and the two priors are mutually singular: Blackwell–Dubins does not apply, and the agents never agree.

7 Applications

7.1 Bayesian learning

The most direct application is to Bayesian inference. Suppose the data $(x_1, x_2, \ldots)$ are drawn from the true measure Q^*. An agent holds a prior π over a family $\{Q_\theta : \theta \in \Theta\}$ of models, inducing a marginal $P = \int Q_\theta \, \pi(d\theta)$. If $P \ll Q^*$ (the true model has positive probability under the prior), then by Theorem 4.1, the agent's posterior predictive distribution merges with Q^* in total variation:

$$\|P(\cdot \mid x_1, \ldots, x_n) - Q^*(\cdot \mid x_1, \ldots, x_n)\|_{\mathrm{TV}} \to 0 \quad Q^*\text{-a.s.}$$

This is a strong form of Bayesian consistency: the agent's predictions about all future events become arbitrarily accurate, regardless of the specific prior, as long as the prior assigns positive probability to the truth.

7.2 Rational expectations and the common prior assumption

In macroeconomics, the merging-of-opinions theorem provides a foundation for the common prior assumption embedded in rational expectations models. If two agents P and Q have mutually absolutely continuous priors and observe a common history, Theorem 4.1 guarantees that they will eventually agree on all conditional forecasts.

Conversely, Aumann [1976]'s agreement theorem shows that agents with a common prior who share their posteriors cannot "agree to disagree" on conditional probabilities. Blackwell–Dubins complements this by showing that even with heterogeneous priors, merging occurs eventually if the priors are in the same absolute-continuity class.

7.3 Signal extraction and diffusion models

Consider the continuous-time signal-extraction model

$$dy_t = \mu_t \, dt + \sigma \, dW_t,$$

where μ_t is an unobserved signal process and W_t is noise. Two agents with different priors over $\{\mu_t\}$ observe the same path $\{y_s : s \leq t\}$. Their respective filtering equations (Kalman–Bucy or Wonham) produce conditional distributions $P(\mu_t|y^t)$ and $Q(\mu_t|y^t)$.

By Theorem 5.4, if their priors over the signal process are mutually absolutely continuous, their filtered distributions merge as $t \to \infty$. The rate of merging is governed by the signal-to-noise ratio: when σ is small, the signal is identified quickly and merging is fast; when σ is large, merging is slow.

7.4 Ergodic Markov chains

For a Markov chain with transition kernel Π and two initial distributions $P_0 = \mu$ and $Q_0 = \nu$, the n-step distributions are $\mu\Pi^n$ and $\nu\Pi^n$. If Π is ergodic with unique stationary distribution π, then both $\mu\Pi^n$ and $\nu\Pi^n$ converge to π, and in particular

$$\|\mu\Pi^n - \nu\Pi^n\|_{\mathrm{TV}} \leq \|\mu\Pi^n - \pi\|_{\mathrm{TV}} + \|\nu\Pi^n - \pi\|_{\mathrm{TV}} \to 0.$$

This is a version of the merging result that does *not* require absolute continuity, because the ergodicity of Π already forces convergence to a common limit. The Blackwell–Dubins theorem is the appropriate generalisation for non-Markovian or non-ergodic processes, where no single invariant measure exists but absolute continuity of the priors is the operative condition.

8 Related Results and Extensions

8.1 Lévy's zero-one law and Doob's consistency theorem

Lévy [1937] established that for any event $A \in \mathscr{F}$,

$$\mathbb{E}[\,\mathbf{1}_A \,|\, \mathscr{F}_n\,] \to \mathbf{1}_A \quad \text{a.s. and in } L^1.$$

This is a precursor to Blackwell–Dubins: it says that *under a single measure*, the conditional probability of any fixed event concentrates on 0 or 1 as information grows. Blackwell–Dubins strengthens this to compare conditional probabilities across two *different* measures.

Doob [1949] proved that, under regularity conditions, the posterior distribution of a parameter θ given data $x_1, \ldots, x_n$ converges to the point mass at the true θ^* for almost every θ^* under the prior. This "Doob consistency" requires the prior to assign positive probability to neighbourhoods of θ^*, which is a form of absolute continuity.

8.2 Diaconis–Freedman and robustness of Bayesian consistency

Diaconis and Freedman [1986] study conditions under which Bayesian procedures are consistent in infinite-dimensional settings. They show that absolute continuity of the prior with respect to the truth is necessary for Doob consistency, connecting their analysis directly to the Blackwell–Dubins framework. They also give examples of priors that are inconsistent when the truth lies outside the support of the prior—a setting in which $P \perp Q$ and Blackwell–Dubins does not apply.

8.3 Rate of merging

Blackwell–Dubins establishes only almost-sure convergence, not a rate. Rates of merging depend on the "distance" between P and Q in a suitable sense. For parametric Bayesian models, the posterior contracts at the $1/\sqrt{n}$ rate (Bernstein-von Mises theorem). For nonparametric models, rates are slower. A useful bound relates the rate of L^1 convergence of Z_n to the rate of merging:

$$\mathbb{E}_Q[d_n] \leq \tfrac{1}{2}\mathbb{E}_Q[|Z_n - Z_\infty|],$$

so any rate for the right-hand side transfers to d_n.

8.4 Merging without absolute continuity

Lehrer and Smorodinsky [1996] and subsequent work in game theory have explored conditions weaker than absolute continuity that still imply merging of predictions on some subsequence of times. These extensions are relevant for the analysis of heterogeneous-agent models in which agents' priors need not be mutually absolutely continuous.

8.5 Summary of the main logical dependencies

The chain of implications underlying the Blackwell–Dubins theorem can be summarised as follows:

$$\begin{aligned} P \ll Q \implies & Z_n = \mathbb{E}_Q[Z_\infty \mid \mathscr{F}_n] \text{ uniformly integrable under } Q \\ \implies & Z_n \xrightarrow{L^1(Q)} Z_\infty \implies d_n \xrightarrow{L^1(Q)} 0 \implies d_n \xrightarrow{Q\text{-a.s.}} 0. \end{aligned}$$

The first implication is the Radon–Nikodym structure. The second is the L^1 martingale convergence theorem. The third is the TV-distance bound (7) together with the supermartingale property. The fourth combines the a.s. convergence of the supermartingale d_n with the L^1 convergence to pin the a.s. limit at zero.

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